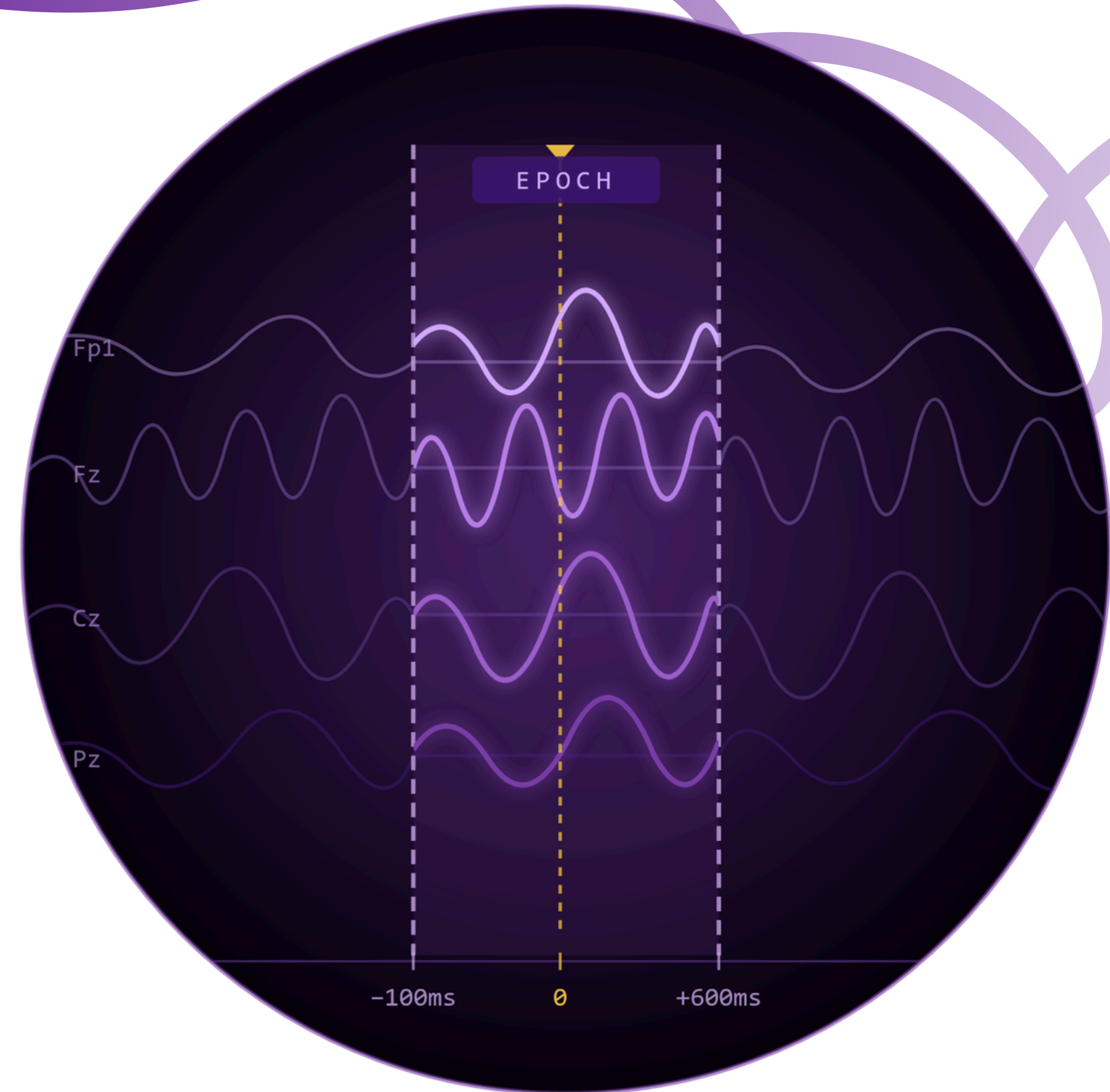
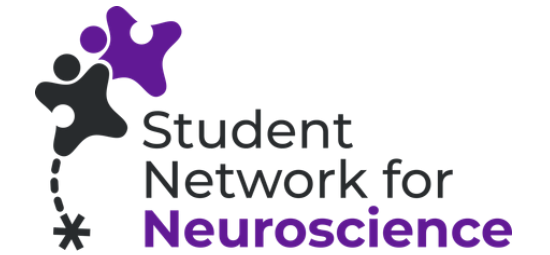


EEG Preprocessing - First steps

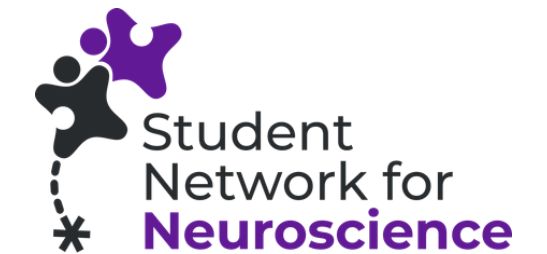


Contents



- Hackathon
- What is EEG?
 - EEG Concept
 - EEG Data format
 - How to access Data
- Preprocessing Pipeline Explained
 - Look into Python and MNE
 - overview of the pipeline steps
- Loading & Merging
- Filtering & Resampling

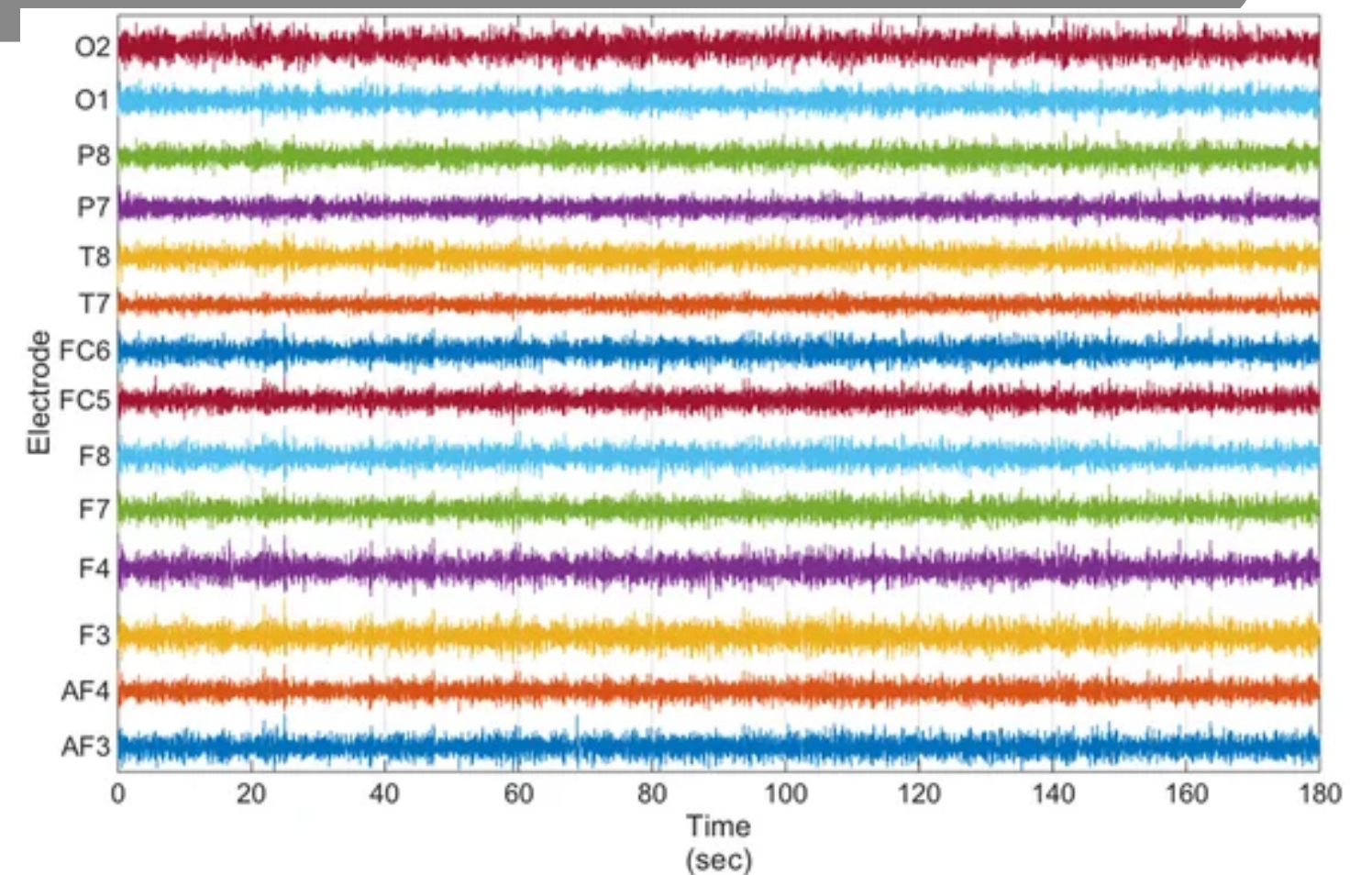
Hackathon



Concept

- Follow-up on our EEG-Workshop
- Learn how to preprocess raw EEG data
 - MNE
 - Python
 - Jupiter Notebook
- Learn some analysis techniques
- Learn to plot and visualize data

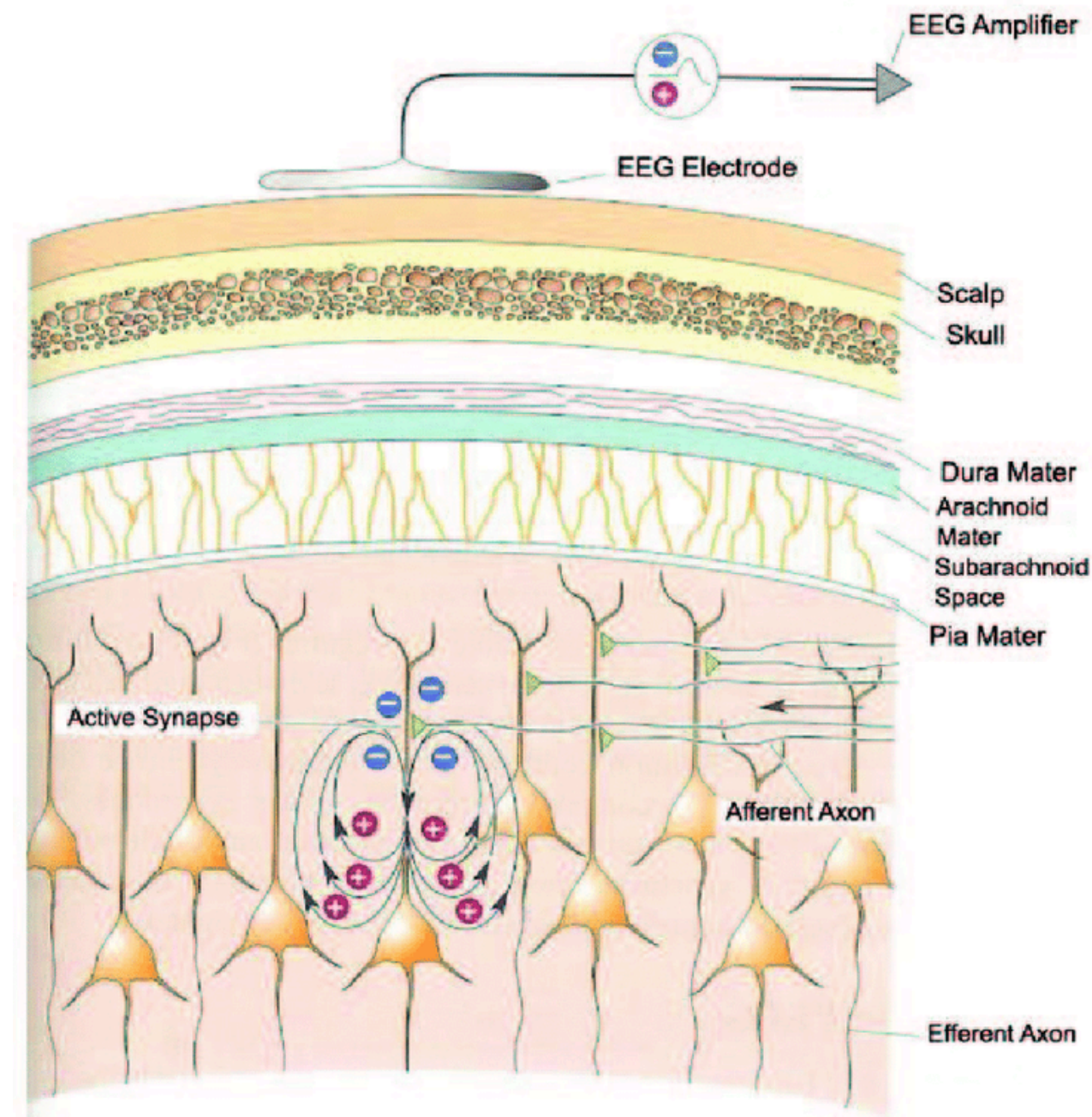
```
# Example for loading a FIF file
raw = mne.io.read_raw_fif('raw.fif', preload=True)
# Example for loading an EDF file
# raw = mne.io.read_raw_edf('my_eeg_data.edf', preload=True)
```



Adapted from Bodranghien et al., Front. Hum. Neurosci., 2017

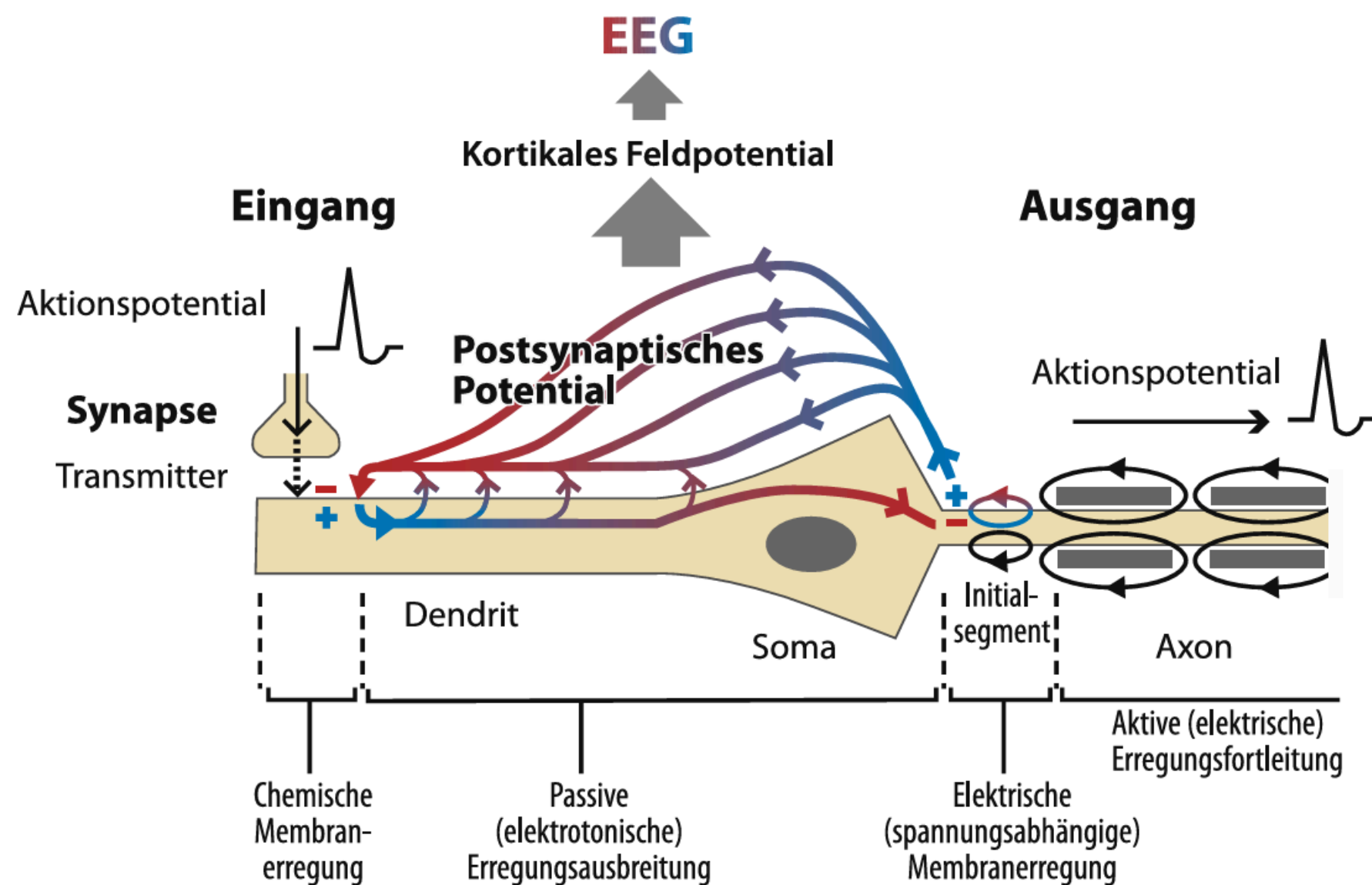
What is EEG?

- Electroencephalography (EEG)
- one of the most important non-invasive methods for studying human cognitive function and diagnosing brain diseases
- mostly measurement of **synaptic** activity of layer 5 pyramidal neurons
- electrodes measure voltage-**differences**(μV)

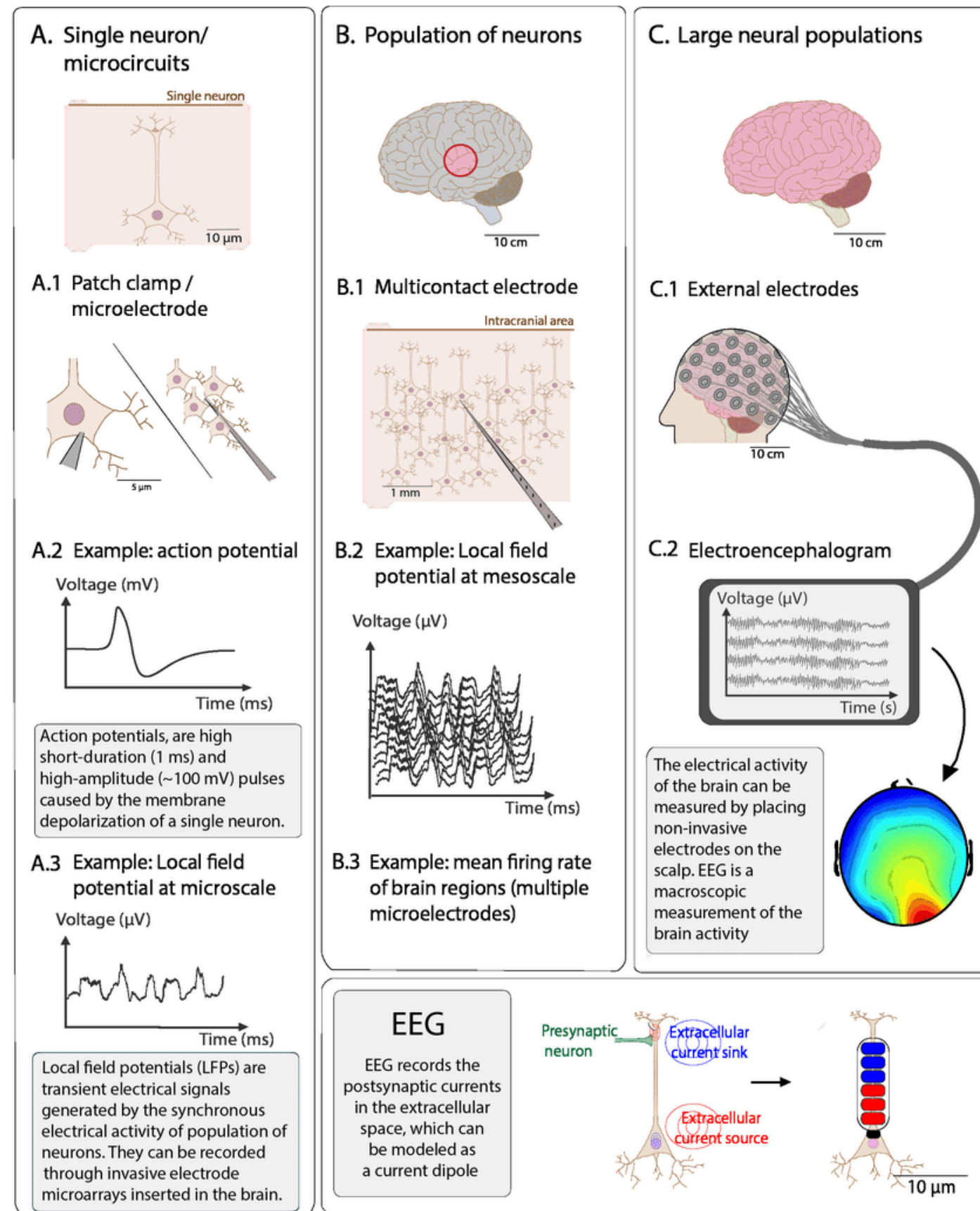


Uldry & Millán, 2007 – Brain–Machine Interface study

What is EEG?



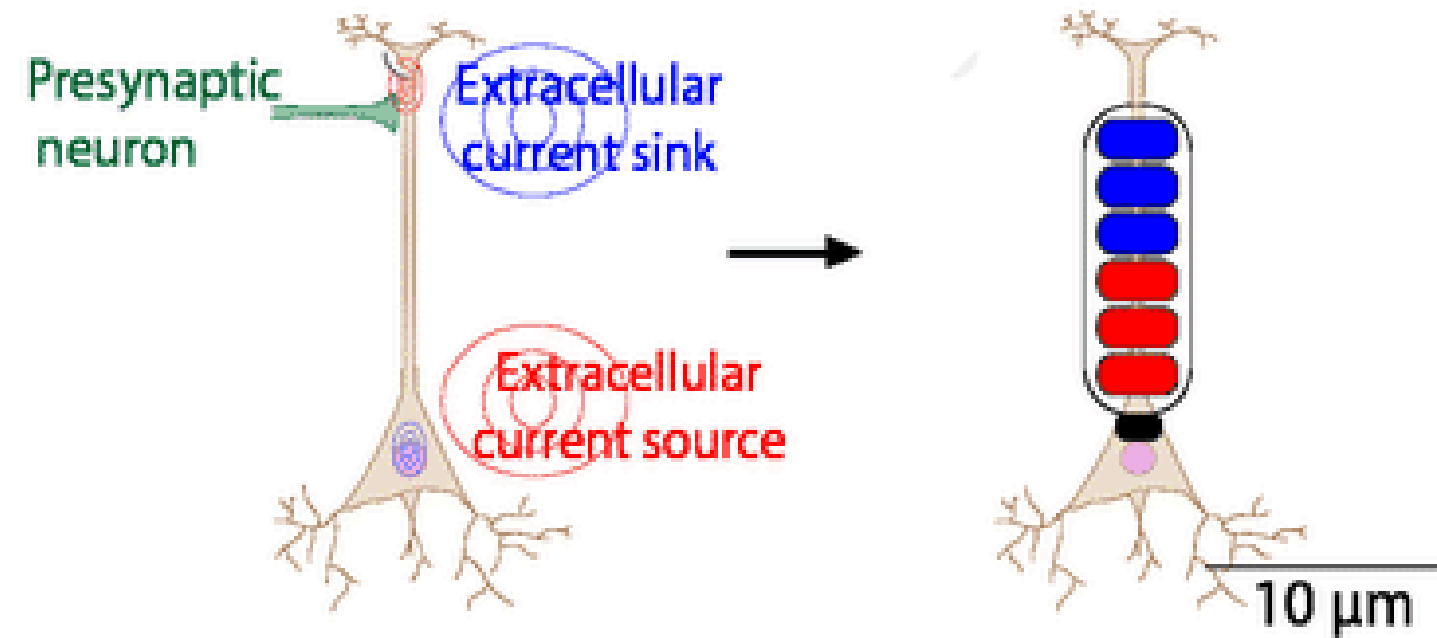
Zschocke, S., & Hansen, H. (2023).



What is EEG?

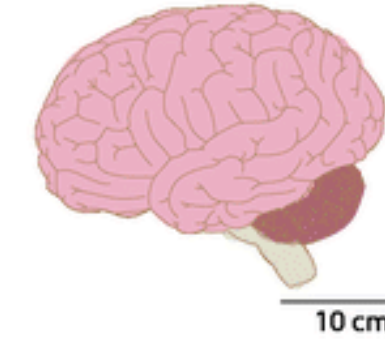
EEG

EEG records the postsynaptic currents in the extracellular space, which can be modeled as a current dipole

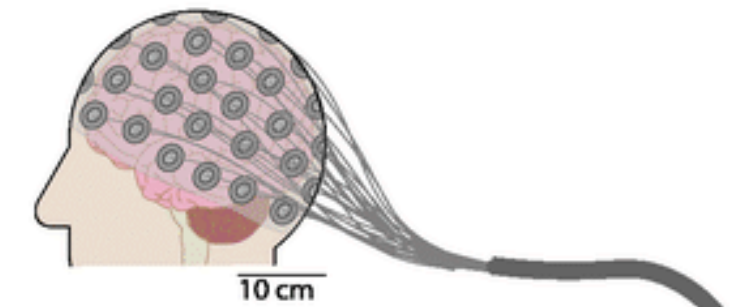


Computational models in Electroencephalography, Glomb, Cabral, Cattani and Mazzoni; September 2020

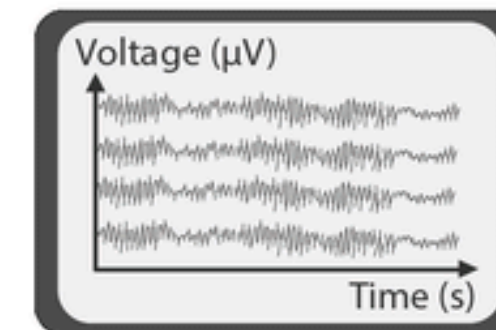
C. Large neural populations



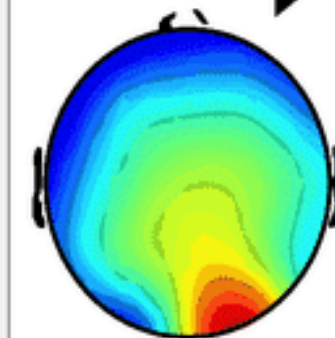
C.1 External electrodes



C.2 Electroencephalogram



The electrical activity of the brain can be measured by placing non-invasive electrodes on the scalp. EEG is a macroscopic measurement of the brain activity



What is EEG?

- Referencing cap position: nasion and inion; left + right pre-auricular points
- Letters:
 - F → frontal
 - Fp → frontopolar
 - T → temporal
 - C → central
 - P → parietal
 - O → occipital
- Numbers:
 - even → right hemisphere
 - odd → left hemisphere
 - middle → z for zero

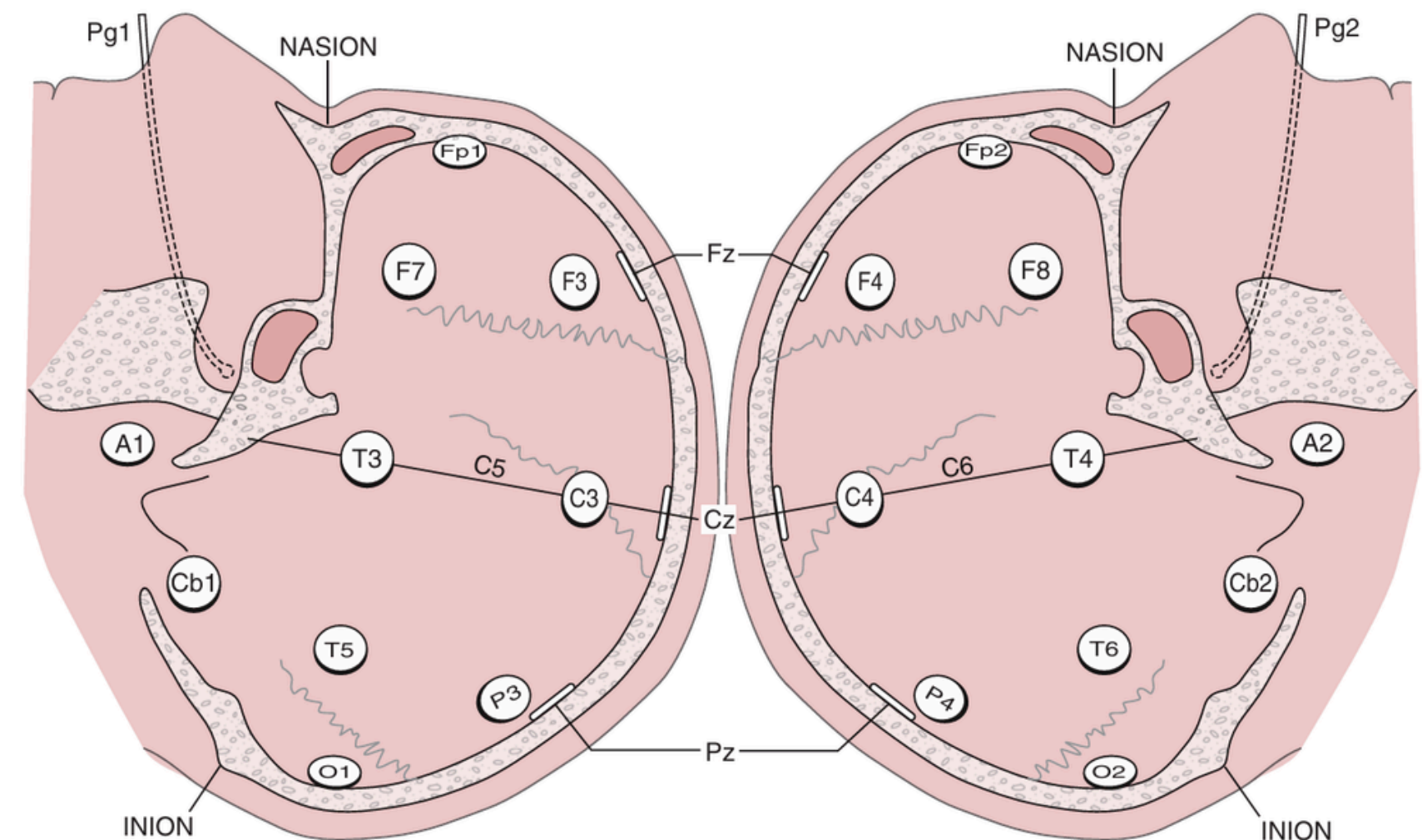


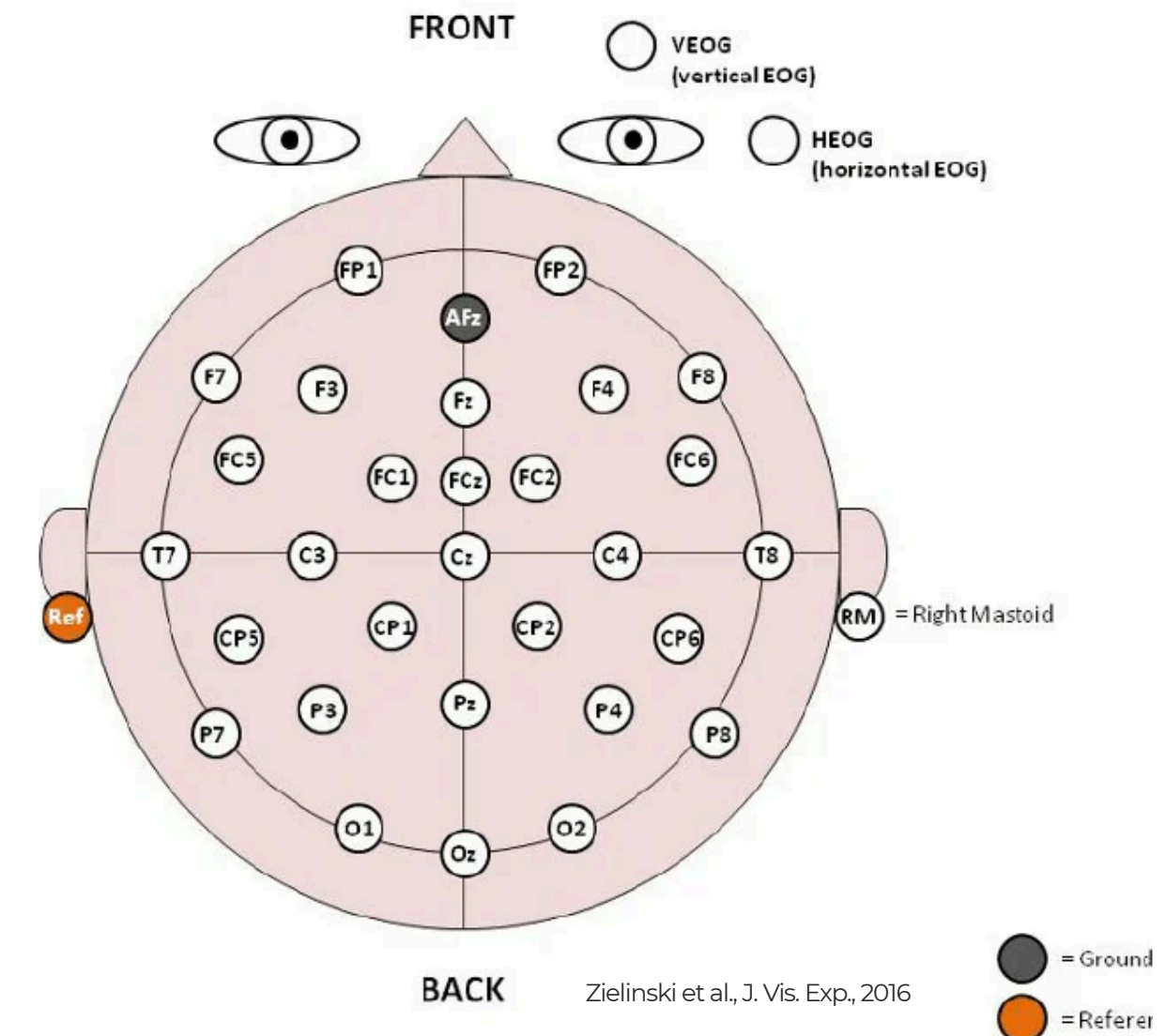
Figure 3.19 These lateral views of the head were based on lateral skull x-rays that Jasper and his colleagues obtained to determine the relationship between the 10-20 electrode positions and the principal fissures of the brain, which are sketched into the figures. (With permission from Jasper 1958, Report on the Committee on Methods of Clinical Examination in Electroencephalography. *Electroencephalography and Clinical Neurophysiology*, 10, 370-375. [https://doi.org/10.1016/0013-4694\(58\)90053-1](https://doi.org/10.1016/0013-4694(58)90053-1).)

What is EEG?

How do reference montage and electrodes setup affect the measured scalp EEG potentials? Hu et. al, 2018

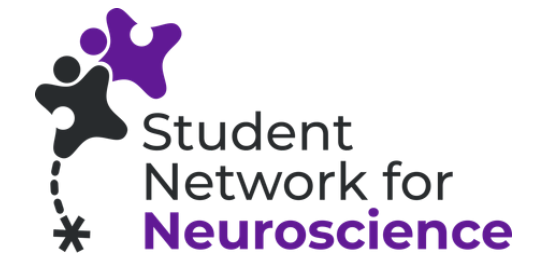
- signal from each recording electrode is measured as a voltage **difference relative** to some reference:
 - (a) single electrode
 - (b) mastoids (M1/M2/Linked)
 - (c) average of electrodes
- ground electrode: typically used to establish a common electrical potential for the recording system
→ minimizes common mode noise
- **Re-referencing** in EEG data analysis: raw recordings with a single reference often contain significant distortions
→ Goal: estimate a more neutral reference point to reduce these effects

$$\mathbf{v}_r = \mathbf{T} \cdot \mathbf{v}_\infty$$



What is EEG?

EEG Concept



Re-Referencing methods:

- Linked Mastoids (LMR)
 - Average Reference (AR): accuracy is influenced by electrode coverage and the orientation of the neural source (dipole orientation)) → For high-density electrode montages, AR can yield better results
 - Reference Electrode Standardization Technique (REST): REST is often considered superior to AR and is less sensitive to perturbations in the head model
 - Bipolar Re-referencing: can reduce spurious correlations between non-adjacent channels more effectively than common average re-referencing (may perform worse for adjacent channels)
- The density of electrodes important:
- higher electrode density → lower re-referencing reconstruction errors for both AR and REST
 - The choice of re-referencing method can significantly impact subsequent analyses, such as network analyses and the localization of seizure onset zones

What is EEG?

Types of Interference Signals (Artifacts) and Filtering Factors

Physiological Artifacts

External/Environmental Artifacts

- Power Line Interference
- Sources of electromagnetic fields in general

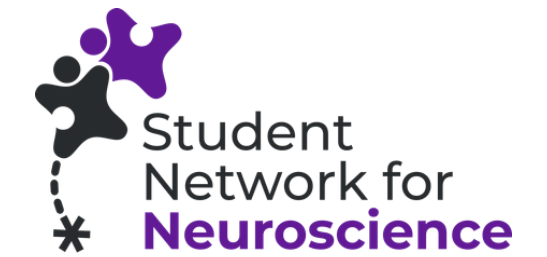
Factors relevant during **filtering**:

- **Signal-to-Noise Ratio (SNR)**
- **Preservation of Neural Information**
- **Online/Real-time Processing** (applications such as Brain-Computer Interfaces (BCI))
- **Adaptive Filtering**
- **Wavelet Transform**

- **Independent Component Analysis (ICA)**
- **Frequency-Spatial Filtering**
- **Head Models and Electrode Positions** (accuracy of positioning)

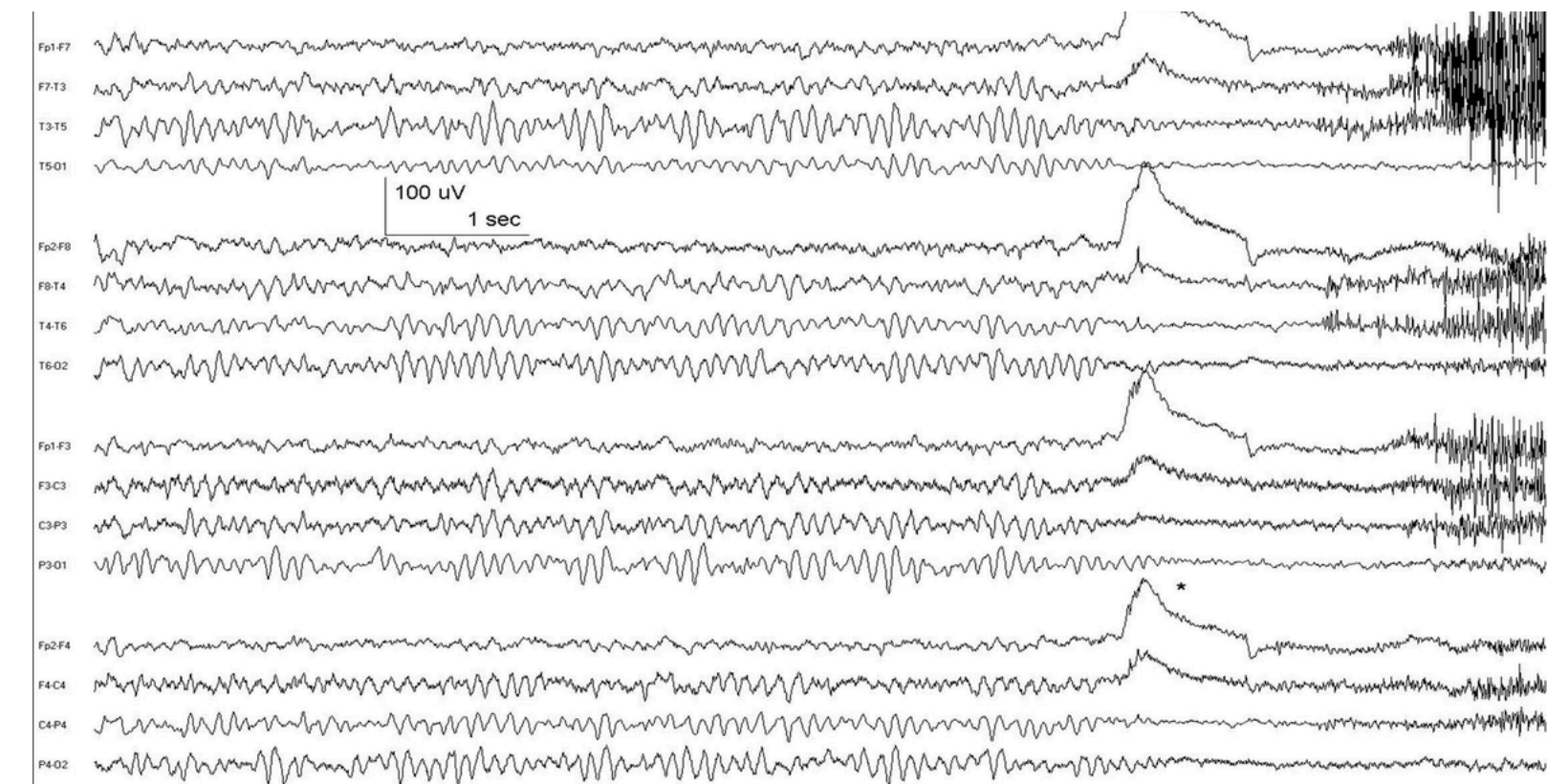
What is EEG?

EEG Data Format



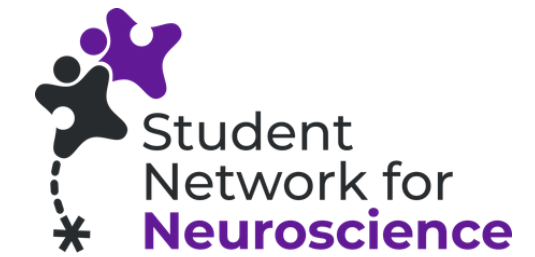
- continuous electrical activity recorded from the scalp over time
- multichannel time series
 - Channels (Electrodes): Each electrode records the electrical potential **difference** to a reference electrode
 - Time Points
 - Amplitude Values
- Associated Metadata
 - Sampling Frequency (Hz)
 - Channel Names and Types
 - Electrode Locations
 - Subject Information
 - Experimental Conditions
 - Event Markers/Annotations

awake, resting; normal posterior alpha rhythm disappears with eye opening (*); high frequency activity after eye opening is muscle artifact; anterior-posterior bipolar montage; 0.5 -70 Hz



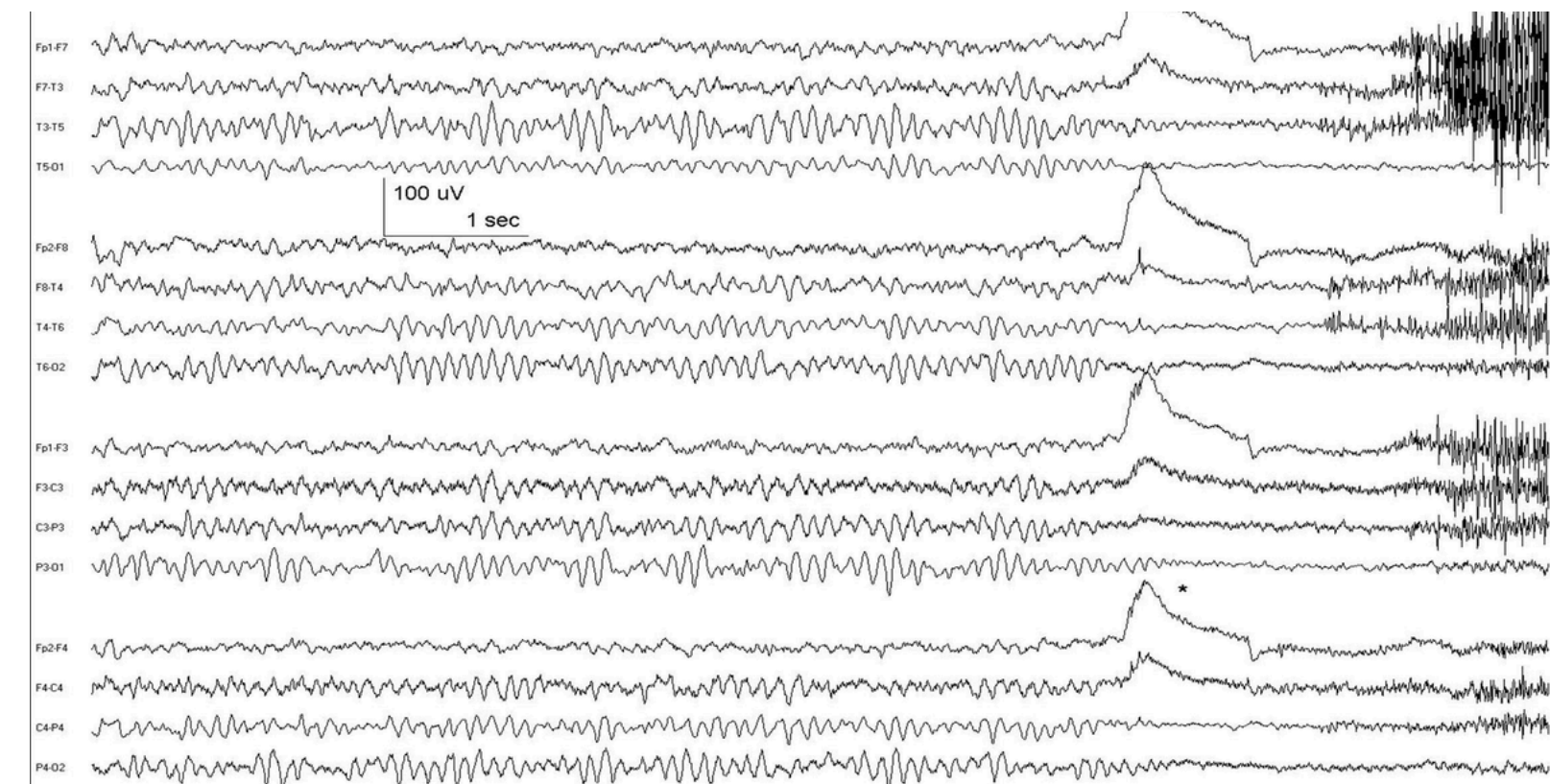
What is EEG?

EEG Data Format



- Common File Formats
 - EDF (European Data Format) / BDF (BioSemi Data Format)
 - BrainVision Core Data Format (VHDR/VMRK/EEG)
 - FIF (MEG/EEG data format)
 - EEG hardware manufacturers' own specific file format
- Standardized Data Structures (BIDS)
 - sub- folders
 - ses- folders
 - eeg- subfolders
 - _events.tsv files

awake, resting; normal posterior alpha rhythm disappears with eye opening (*); high frequency activity after eye opening is muscle artifact;
anterior-posterior bipolar montage; 0.5 -70 Hz



What is EEG?

How to access the data

- Usually: Some sort of programming environment (e.g. Python)
- MNE-Python: open-source software package
- But also: User-Friendly Interfaces
 - tools like EEG-Pype provide a graphical user interface (GUI) built on top of MNE-Python functions
 - Similarly, other toolboxes like osl-ephys also build upon MNE-Python
 - toolboxes like EEGLAB provide an alternative standalone processing package

```
import os

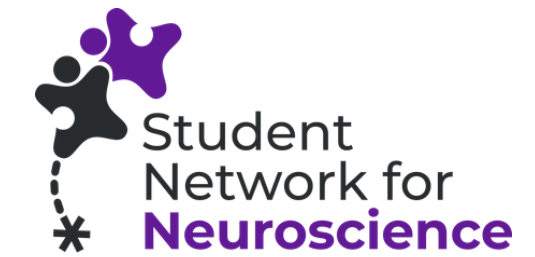
import numpy as np

import mne

sample_data_folder =
mne.datasets.sample.data_path()
sample_data_raw_file = os.path.join(
    sample_data_folder, "MEG", "sample",
    "sample_audvis_raw.fif"
)
raw =
mne.io.read_raw_fif(sample_data_raw_file)
raw.load_data()
```

Programmes

Python



- open-source coding language
- object-oriented, high-level programming language with dynamic semantics
- widely used in web applications, software development, data science, and machine learning (ML)
- quick and easy to learn and understand
- further introduction to Python is in the GitHub repository

A screenshot of an IDE window titled "Introduction to Python". The main editor shows a file named "hello_world.py" with the following code:

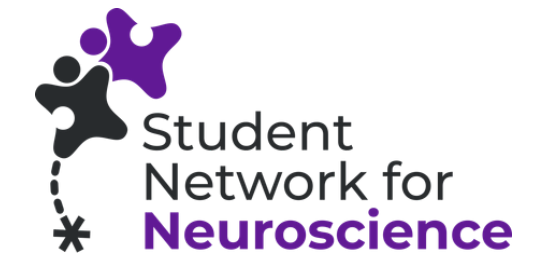
```
1 print("Hello, world! My name is NAME")
2
```

The Python Console at the bottom shows the command prompt and the execution of the code. The console output is:

```
"C:\Users\r...\.idea\VirtualEnvironment\Scripts\python.exe"
import sys; print('Python %s on %s' % (sys.version, sys.platform))
sys.path.extend(['C:\Users\r...'])
>>> Python Console
>>>
```

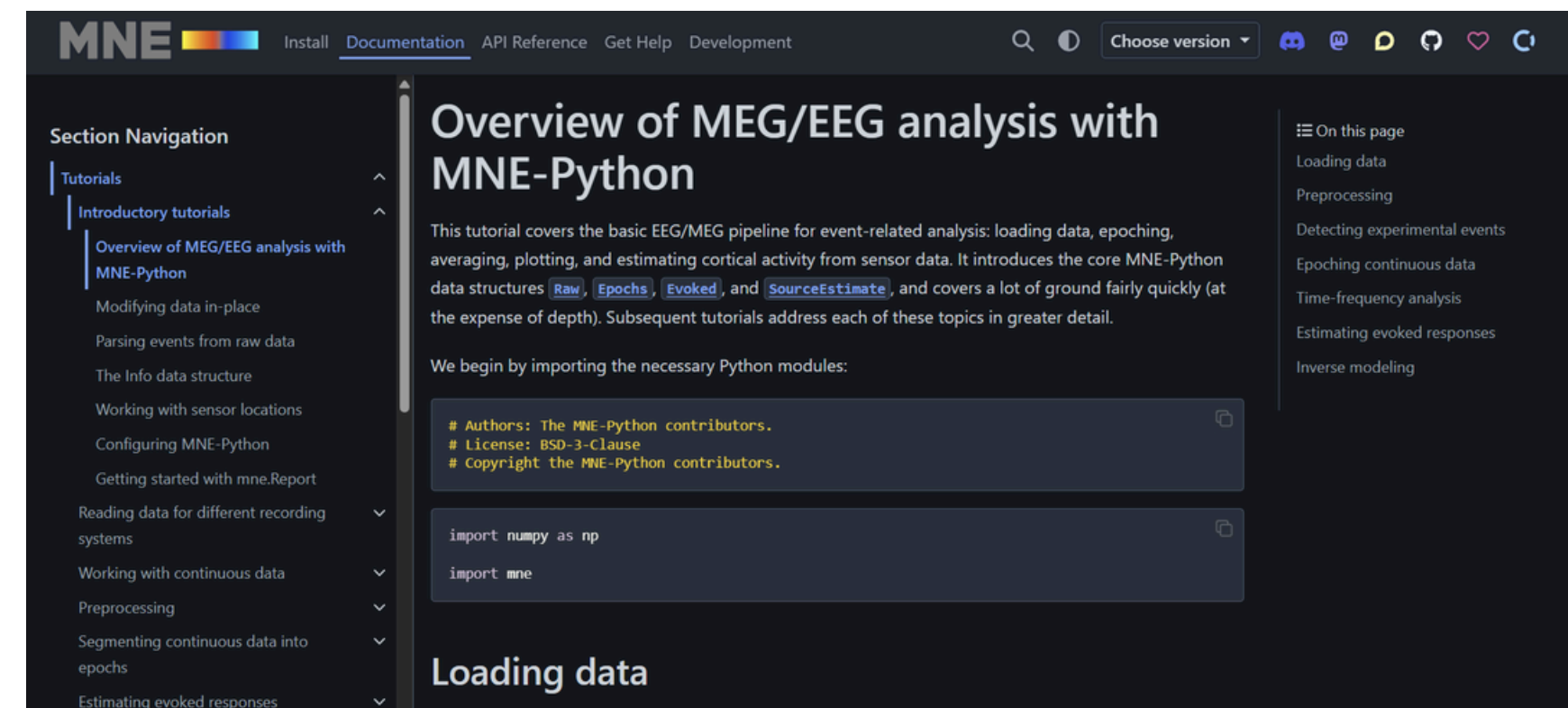
The status bar at the bottom indicates the file is "hello_world.py", the encoding is "UTF-8", and the Python version is "Python 3.6 (Introduction to Python)".

Programmes

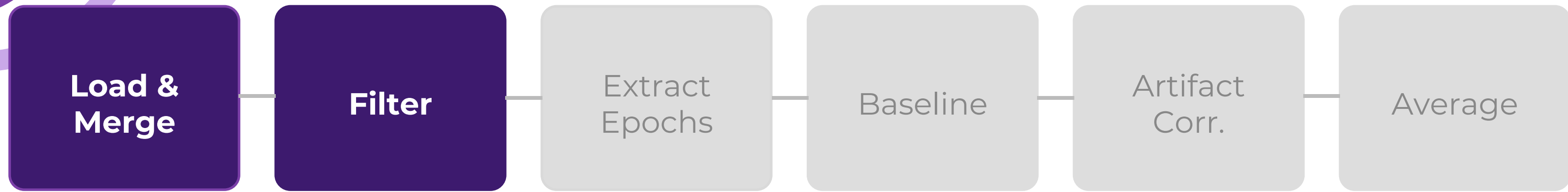


MNE

- open-source Python software
- processing, analyzing, and visualizing electroencephalography (EEG) and magnetoencephalography (MEG) data
- comprehensive tools for the full analysis pipeline
 - pre-processing (filtering, artifact removal)
 - sensor-level analysis
 - source estimation
 - time-frequency analysis
 - even machine learning approaches (encoding/decoding)
 - visualization and plotting



Pipeline



Today: Steps 1 & 2

- Today's tasks:
 - Load data into Python
 - Merge data
 - Filter and resample data
- along the way: visualize your progress

Load & Merge Data

Used file format: BrainVision Core Data Format

- **.vhdr:** Header file, metadata about EEG recording; text-based file with information (sampling rate, number of channels, channel names, electrode impedances,... and **references** to corresponding .vmrk and .eeg files)
 - primary entry point for software
- **.vmrk:** Marker file, stores event markers or annotations for events/stimuli; essential for segmenting the continuous EEG data into epochs
- **.eeg:** Data file; contains the raw, continuous EEG time-series data; typically a binary file that stores the actual voltage values recorded from each electrode at each time point

Load Data

To load the files into MNE-Python, you need to specify the path to the .vhdr file
→ Use read_raw_brainvision function to automatically locate and read the associated .vmrk and .eeg files

```
import mne
# Define the path to your .vhdr file
vhdr_file_path = 'path/to/your/eeg_data.vhdr'
# Load the BrainVision data
raw = mne.io.read_raw_brainvision(vhdr_file_path,
preload=True)
# Print some information about the loaded data
print(raw.info)
print(raw.ch_names[5]) # Print first 5 channel names
print(raw.get_data().shape) # Print data shape (channels, times)
# You can then proceed with further preprocessing and
analysis
# For example, to plot the raw data:
# raw.plot(duration=10, n_channels=30, scalings='auto')
```

- `mne.io.read_raw_brainvision()` is the function used to read BrainVision files.
- `vhdr_file_path` should be replaced with the actual path to your .vhdr file.
- `preload=True` loads all the data into memory, which is often convenient for further processing. If your dataset is very large, you might set `preload=False` and load data in chunks.
- The raw object returned is an MNE-Python Raw object, which contains the EEG data, channel information, and event markers.

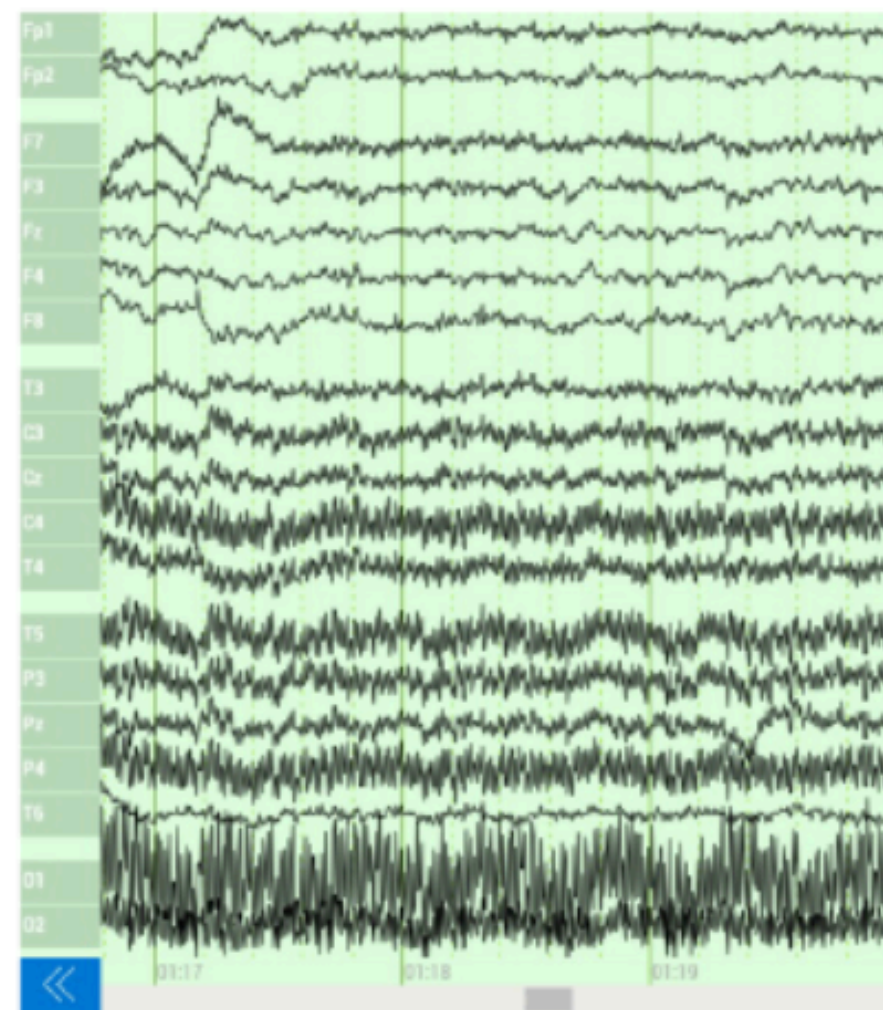
Merge Data

Merging can be done by using the ‘concatenate’ functionality

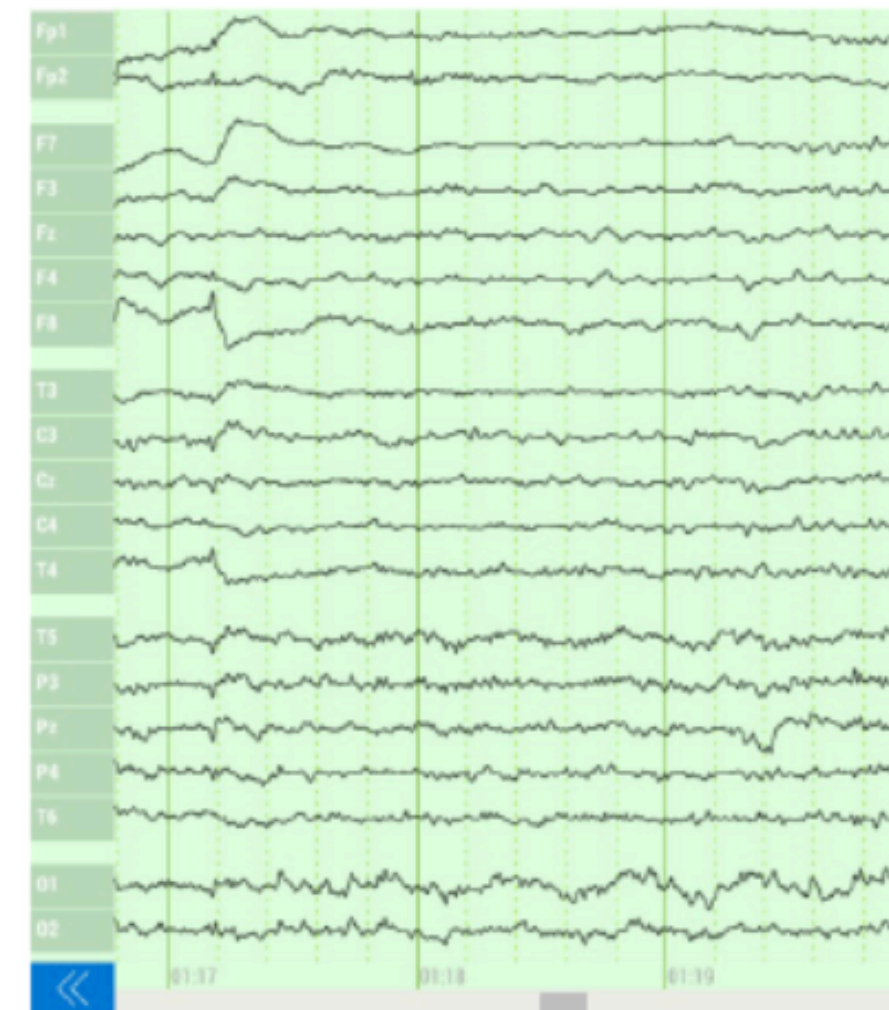
- there is a different function for each object type in MNE (raw, epochs, events)
- datasets to be merged need to be preloaded
- at the merging site, a ‘BAD boundary’ annotation is inserted

Filtering

Filtering removes or attenuates parts of a signal. Specifically it attenuates frequencies.



Unfiltered EEG Signal

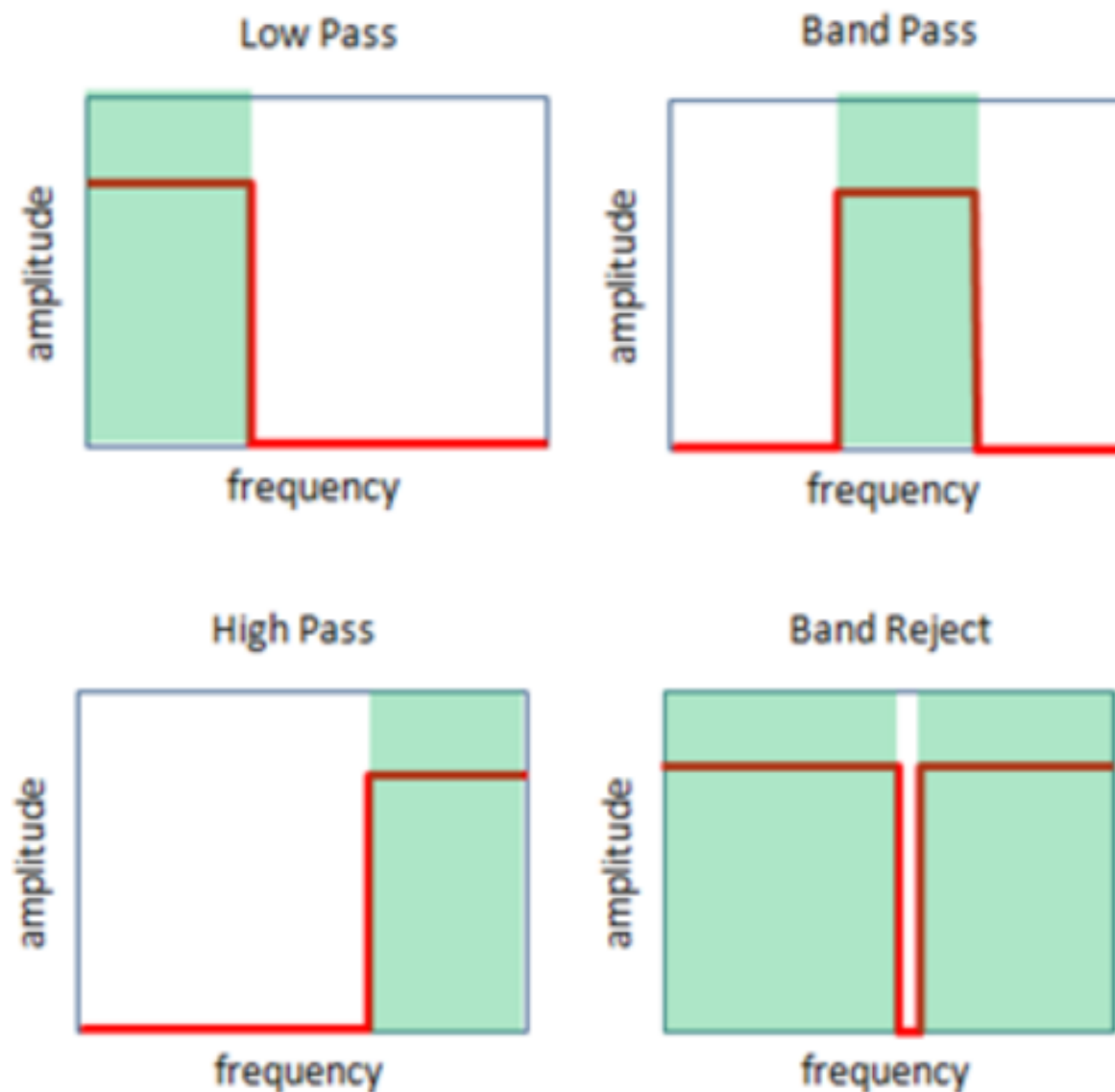


Filtered EEG Signal

<https://zeto-inc.com/blog/eeg-signal-enhancement-digital-eeg-filters/#:~:text=Three%20main%20types%20of%20filters%20used%20to%20accomplish,also%20called%20notch%20filters%29%20%E2%80%93%20See%20Figure%201.>

Filtering

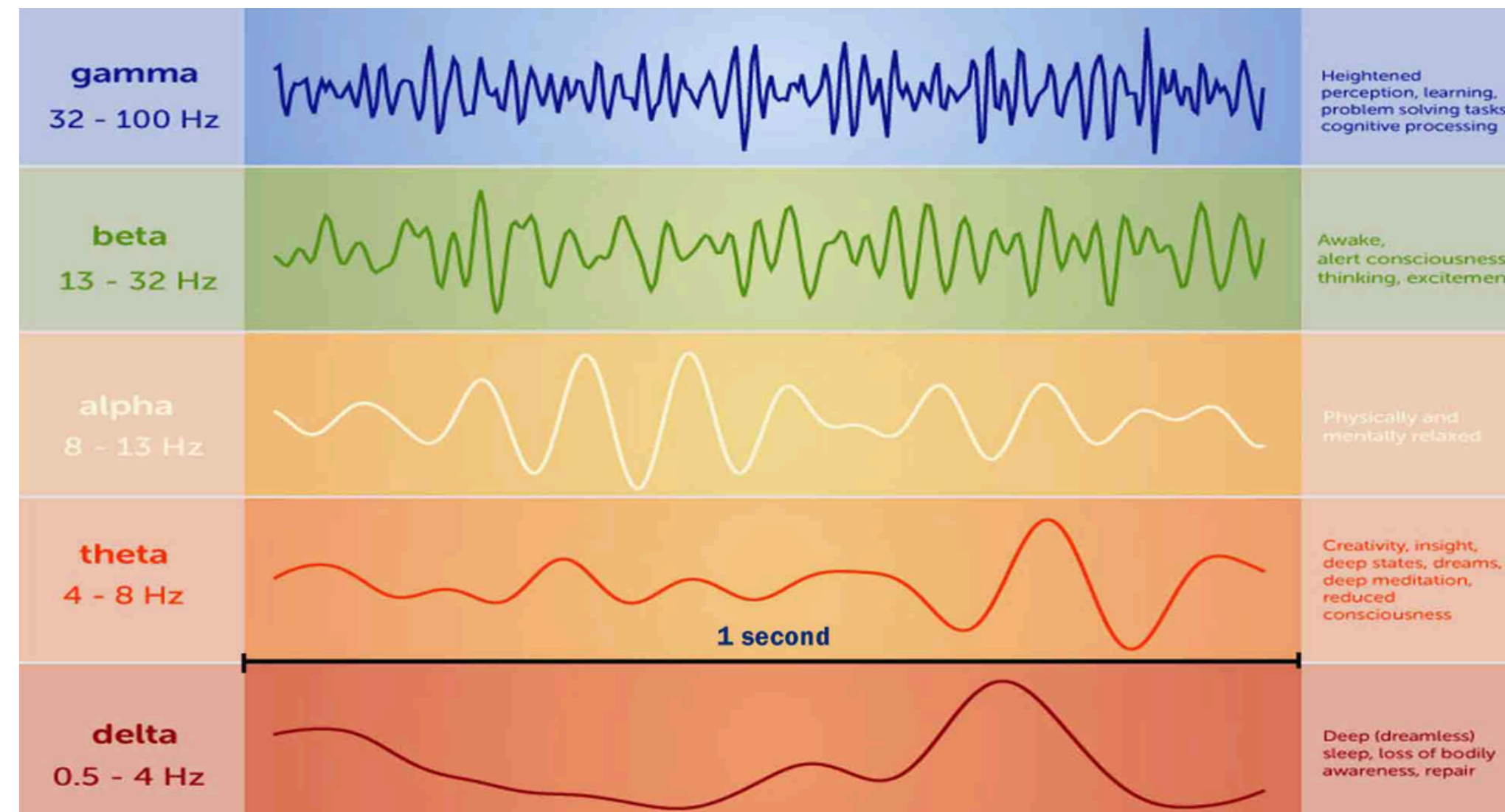
There are many types of filters that only remove pre-specified frequencies.



- filter types can help with the removal of typical recording artifacts
- mainly frequencies that are not reflecting neuronal oscillations
- Reminder: 1 Hz = one cycle per second

Filtering

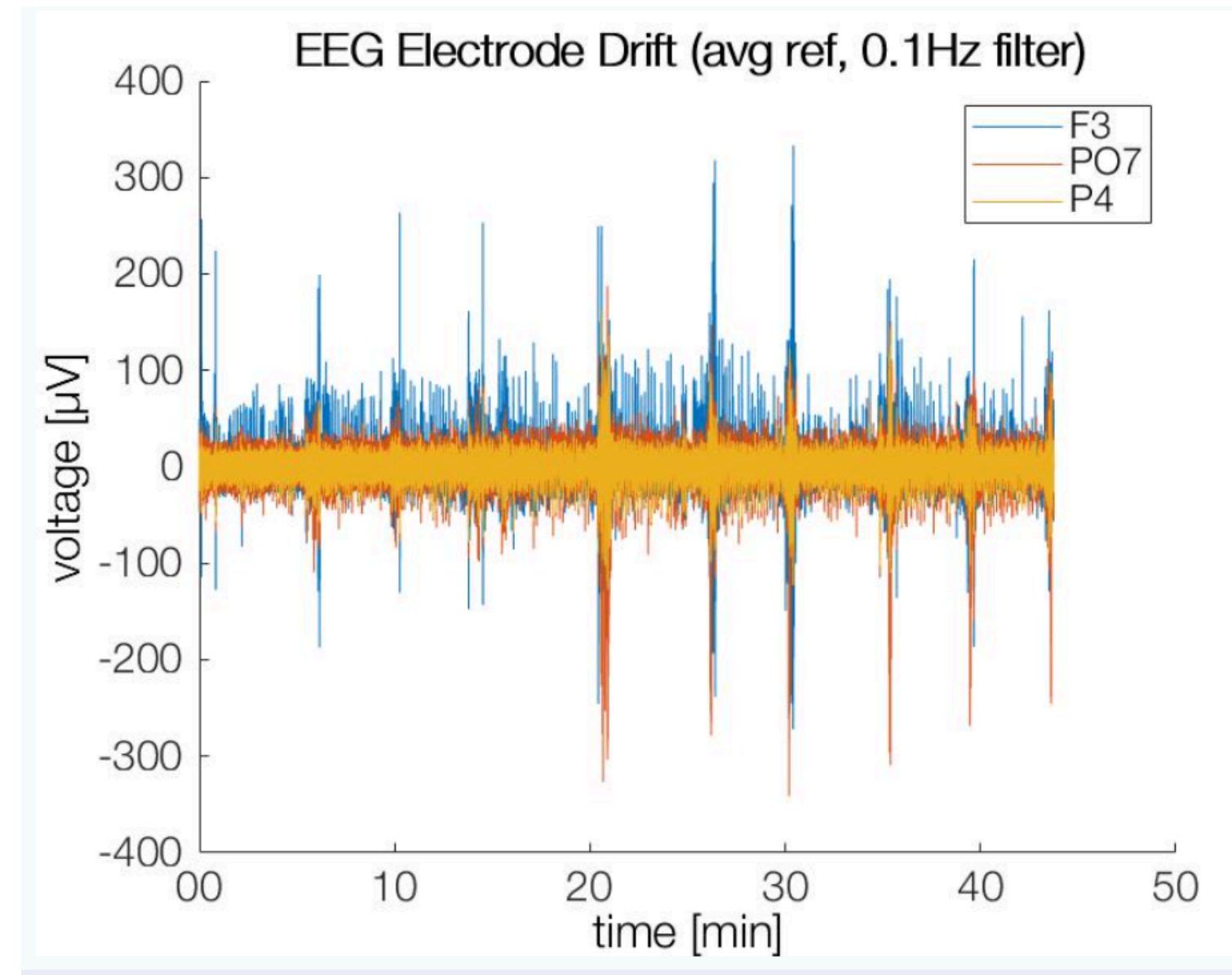
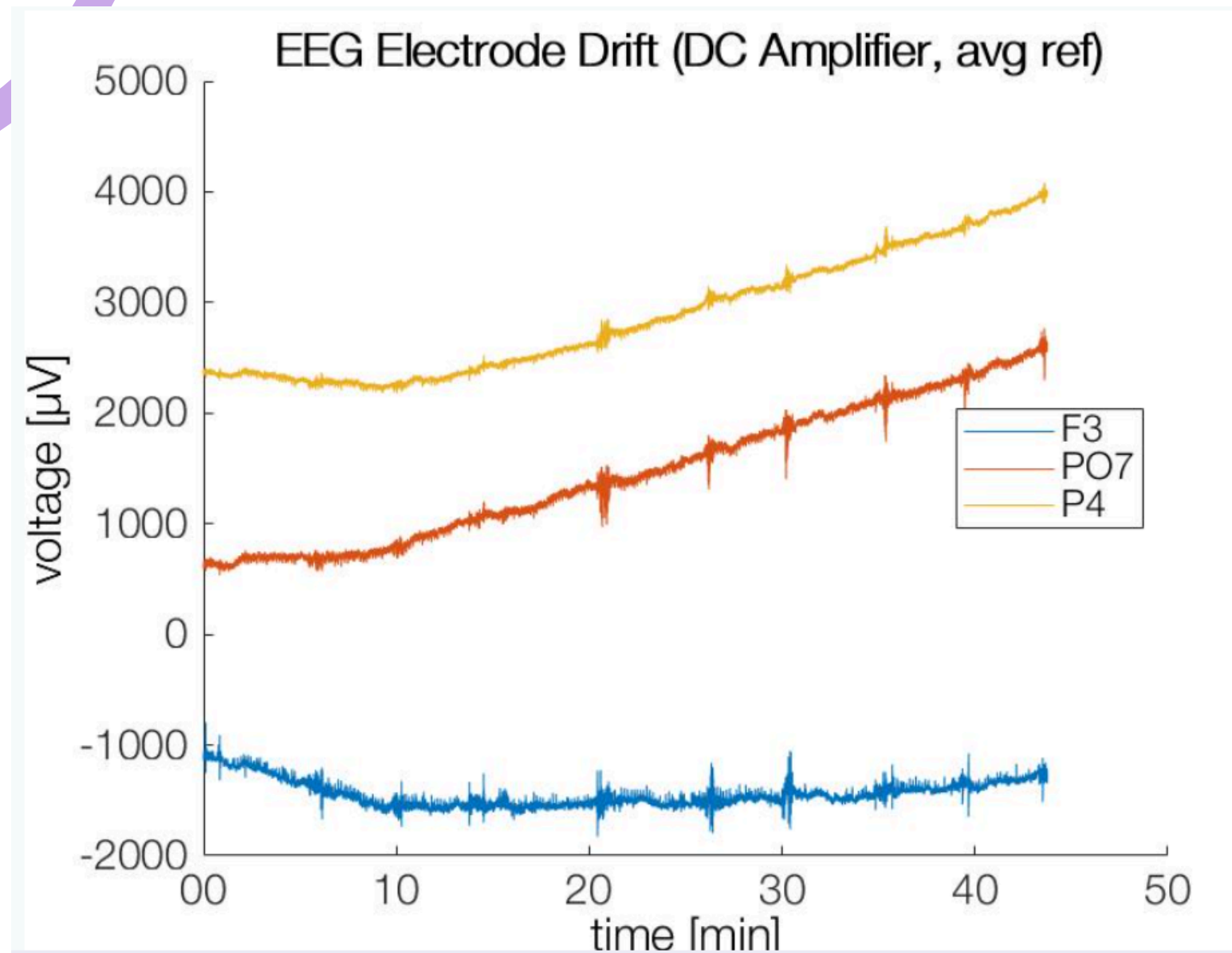
**Not all oscillations are originating from the brain.
And from the ones that are, not all might be relevant for your paradigm.**



<https://biofeedback-neurofeedback-therapy.com/brainwaves-in-neurofeedback/>

Filtering

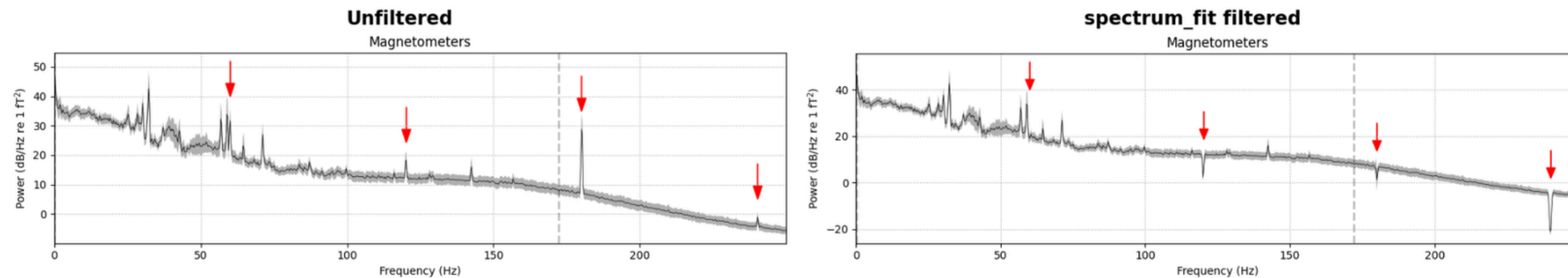
Filtering of low frequency drifts (e.g. sweat artifacts)



<https://benediktehinger.de/blog/science/electrode-drift-in-eeg/>

Filtering

Filtering of power line noise



https://mne.tools/stable/auto_tutorials/preprocessing/30_filtering_resampling.html

- power line noise mainly in MEG, since EEG is mostly done in shielded chambers
- frequencies:
 - 50 Hz in Europe
 - 60 Hz in the US
 - other countries may differ
- also affects all the respective harmonic frequencies
- There is so much more to filtering...but this is just to illustrate the concept!

Filtering

How could the code look like?

1. High-pass filter

```
raw_highpass = raw.copy().filter(l_freq=1.0, h_freq=None)
```

2. Low-pass filter

```
raw_lowpass = raw.copy().filter(l_freq=None, h_freq=40.0)
```

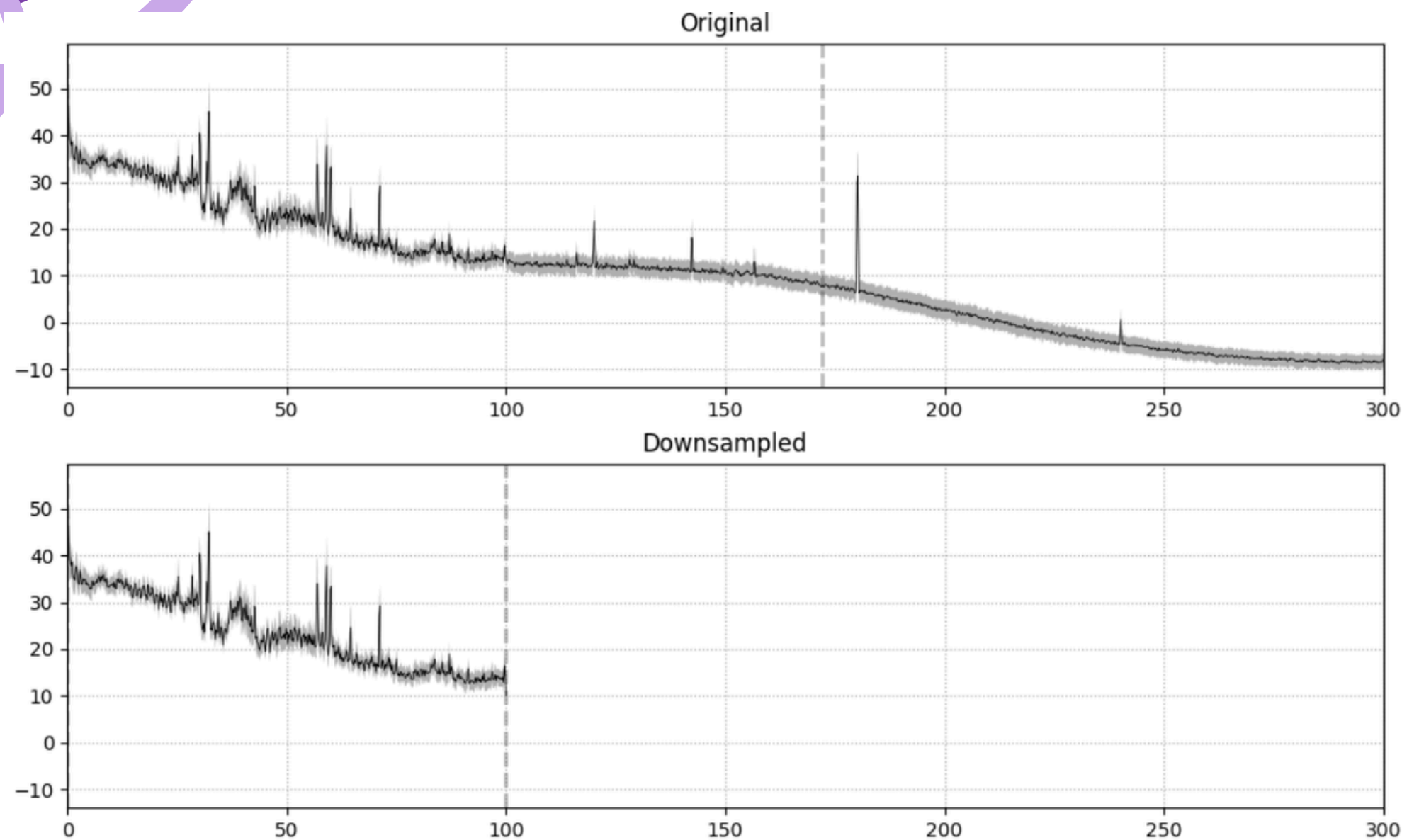
3. Band-pass filter

```
raw_bandpass = raw.copy().filter(l_freq=1.0, h_freq=40.0)
```

- three filter examples
- avoid the usage of multiple filters for the same data as that will lead to artifacts
- ‘.copy()’ is only useful if you want your original data untouched
→ be mindful with that, since it can easily overload your memory if used repeatedly
- Which filter do we need in our script?

Resampling

Works similarly to filtering, but completely removes specified frequencies.



- Downsampling saves computation time and memory space
- no information is lost, but timing of events can shift slightly
 - therefore event codes **MUST** be resampled as well to ensure precise timing
- resampling incorporates a low-pass filter per default to avoid aliasing
 - aliasing = overfolding of higher frequencies into lower ones during resampling
- double filtering is avoided if we used a smaller low-pass filter before

https://mne.tools/stable/auto_tutorials/preprocessing/30_filtering_resampling.html

Resampling

How do I show all these spectral plots of frequencies?

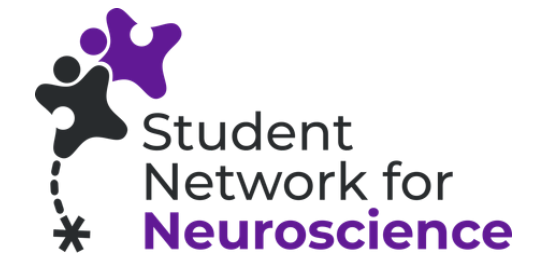
```
# resample our raw dataset
raw, events = raw.resample(sfreq=256, events=events[0])

# look at the new sampling frequency
print(f'New sampling frequency: {raw.info["sfreq"]} Hz')

# we can also look at the power spectral density
spectrum_resampled = raw.compute_psd()
spectrum_resampled.plot(average=True, picks='data',
exclude='bads', amplitude=False)
```

- specify the frequency to which you want to downsample
 - Hint: everything to the power of 2 saves computation time (and is more precise mathematically)
- before plotting the power spectrum, calculate it with `'compute_psd()'`

Save the data



How do I finally save my data as a .fif file for further usage?

Can you figure that out yourself?

Always keep in mind to look at MNEs official documentation!

<https://mne.tools/stable/documentation/index.html>

Do you have any questions?

Happy Hacking!

**The task sheets and EEG data sets are in
the GitHub repository.**

Überschrift

